

C.3

Wave phenomena

Guiding questions

- How are observations of wave behaviours at a boundary between different media explained?
- How is the behaviour of waves passing through apertures represented?
- What happens when two waves meet at a point in space?

What are the basic behaviours of all waves?

- Reflection
- Refraction
- Diffraction
- Interference

Each of these properties will be discussed in this topic. But first we need to consider how we can represent travelling waves in two dimensions on paper, or screens.

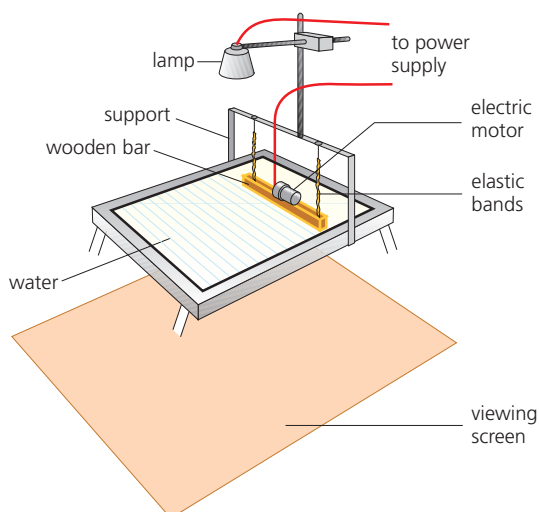
Wavefronts and rays

SYLLABUS CONTENT

- ▶ Waves travelling in two and three dimensions can be described through the concepts of wavefronts and rays.

◆ **Ripple tank** A tank of shallow water used for investigating wave properties.

◆ **Wavefront** A line connecting adjacent points moving in phase (for example, crests). Wavefronts are one wavelength apart and perpendicular to the rays that represent them.



■ **Figure C3.1** A ripple tank is used to investigate wave behaviour

Waves in two dimensions

Figure C3.1 shows a **ripple tank**: a common arrangement used to observe the behaviour of waves. Small waves (ripples) can be made by touching the surface of shallow water at a point, or with a wooden bar. Usually, a motor is attached to the bar to make it vibrate and produce continuous parallel waves at various frequencies. The light above enables the moving waves to be seen on the screen below the tank. (A *stroboscope* is often used to make the waves appear stationary.)

The ‘waves’ seen on the screen show the positions of wave *crests*. These lines are called **wavefronts**. They are one wavelength apart. More precisely:

A wavefront is a line joining neighbouring points moving in phase with each other.

The blue lines seen in Figure C3.2 represent the pattern produced by regular disturbances of the water at point P. The waves are spreading with equal speed in all directions (in two dimensions), so that the pattern is circular. Wave speed depends on the depth of the water, which is usually constant if the tank is horizontal.

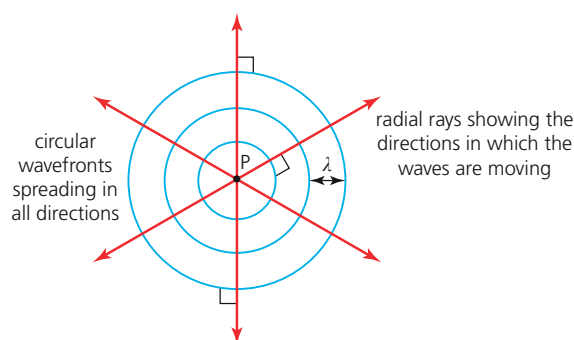
◆ **Ray** A line showing the direction in which a wave is transferring energy.

◆ **Radial** Diverging in straight lines from a point.

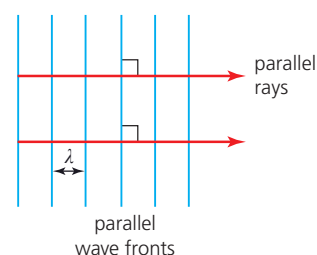
The red lines with arrows in Figure C3.2 are called **rays**. These rays can be described as **radial**, meaning that they are spreading out in straight lines from a point. Radial rays represent circular (or spherical) wavefronts.

Rays are lines showing the direction in which wavefronts are moving and energy is transferred. Rays and wavefronts are always perpendicular to each other.

The wavefronts that we will be considering in this course are either circular (or spherical), see Figure C3.2, or the other common possibility: parallel wavefronts, as shown in Figure C3.3. Parallel wavefronts can be made on the water surface in a ripple tank by using a wooden bar.



■ **Figure C3.2** Circular wavefronts and radial rays spreading from a point source



■ **Figure C3.3** Parallel wavefronts and parallel rays that are not spreading out

The movement of parallel wavefronts is represented by parallel rays.

Waves in three dimensions

Waves spreading from a point with constant speed in three dimensions can be represented by spherical wavefronts. If the source of waves is a very long way away (compared to the wavelength) the wavefronts will be (almost) straight and parallel. Such waves are described as **plane waves**. A common example: light waves from a distant point. The Sun is an obvious source of plane waves.

◆ **Plane waves** Waves travelling in three dimensions with parallel wavefronts, which can be represented by parallel rays.

◆ **Visualization** Helping understanding by using images (mental or graphic).



Knowledge and the knower, The natural sciences

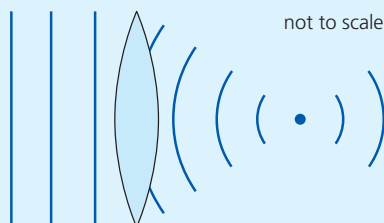
- How do our interactions with the material world shape our knowledge?
- What is the role of imagination and intuition in the creation of hypotheses in the natural sciences?

Different ways of describing the same thing

We can use either the concept of wavefronts, or the concept of rays, to describe the movement of the same waves, but they are not normally used at the same time.

The use of wavefronts to describe waves on water surfaces, which are easily *visible*, is easily understood, but the associated concept of ‘rays’ seems unnecessary. However, when describing another wave phenomenon, the *invisible* passage of light through the air, why do we usually prefer the **visualization** of rays, although they have no physical reality? How might such visualizations extend, affect, or perhaps limit our understanding of the natural world?

- 1 a Describe and explain how parallel wavefronts in a ripple tank appear if they are moving perpendicularly into shallower water.
 b Discuss how the wavefronts from a point source would appear if the tank was raised on one side. Explain your answer.
- 2 Figure C3.4 represents some light wavefronts passing from left to right through a lens.



■ **Figure C3.4** Light wavefronts passing from left to right through a lens

- a Explain how you know that the source of light is a long way away.
- b Make a sketch of the lens and show the path of five rays to represent the movement of the wavefronts.

- c State which word we use to describe the effect of the lens on the light rays.

- 3 Figure C3.5 shows ocean waves as they approach the coast. Suggest possible reasons why the separation and direction of the waves change.



■ **Figure C3.5** Ocean waves refracting (and diffracting) as they approach a beach

■ Transmission, absorption and scattering of waves

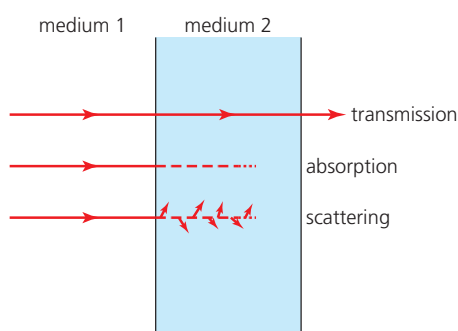
SYLLABUS CONTENT

- Wave behaviour at boundaries in terms of transmission.

◆ **Transmission** Passage through a medium without absorption or scattering.

◆ **Opaque** Unable to transmit light (or other forms of energy).

The process in which waves are able to travel through a medium is described as wave **transmission**. For example, light and sound can be transmitted through air and water. During the transmission of waves, there is often *absorption* of energy: some or all of the wave energy is transferred to internal energy within the medium. Waves may also be randomly misdirected by interactions with irregularities within the medium. This is called scattering. Figure C3.6 illustrates these terms: waves are transmitted by medium 1, then enter medium 2, where they are each either transmitted, absorbed or scattered. In reality, all three processes can occur with the same waves.



■ **Figure C3.6** Transmission, absorption and scattering

A medium through which light can be transmitted, and through which we can see clearly, is described as being *transparent*. A medium through which light cannot be transmitted is described as **opaque**.

Wave power, intensity and amplitude

As waves spread out, and/or their energy is dissipated, the power that they transfer is reduced. We usually describe this as a reduction of wave intensity, a concept that was introduced in Topic B.1, and is defined again here:

$$\text{intensity, } I = \frac{P}{A} \quad \text{SI Unit: } \text{W m}^{-2}$$

Common mistake

Some books use the symbol A to represent amplitude, but this can cause confusion with the symbol for area. We will use the symbol A to represent area and the symbol x_0 to represent amplitude.

Speaking generally, we know that the energy transferred by a wave is proportional to its amplitude squared (Topic C.2). More precisely:

$$\text{intensity} \propto \text{amplitude}^2$$

Waves spreading from a point without absorption

There are reasons, discussed above, why waves may lose energy during transmission through a medium, but, if the waves were *spreading* out from a point source (that is, they are not plane waves), their intensity will decrease for that reason alone, without any absorption.

In two dimensions (surface waves)

As surface waves spread away from a point source the wavefronts will extend over greater and greater lengths. For example, if a circular wavefront increases its distance from its centre from r to $2r$, then its circumference will increase from $2\pi r$ to $4\pi r$. See Figure C3.7.

In each spreading wavefront the same amount of energy is spread over a greater length, so that the wave amplitude will decrease. If surface waves are a great distance from their source, the wavefronts will become almost parallel to each other, so that no loss of energy / power may be noticeable.

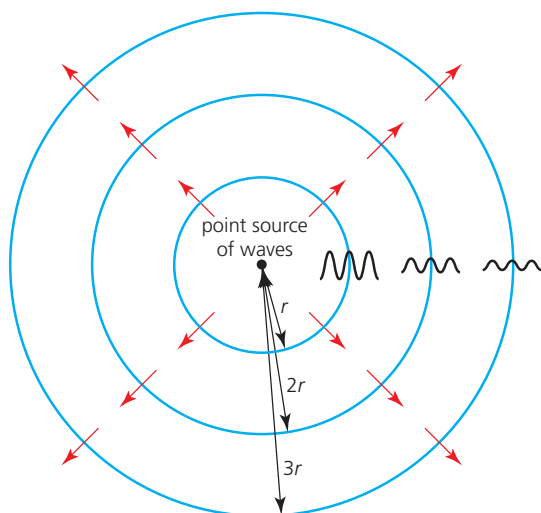
In three dimensions (such as light and sound)

We discussed the spreading of light and infrared waves from the Sun in Topic B.2. Refer back to Figure B2.1 in that chapter. The intensity of any waves spreading radially in three dimensions, without absorption, from a point source follows an inverse square relationship, which is repeated here:

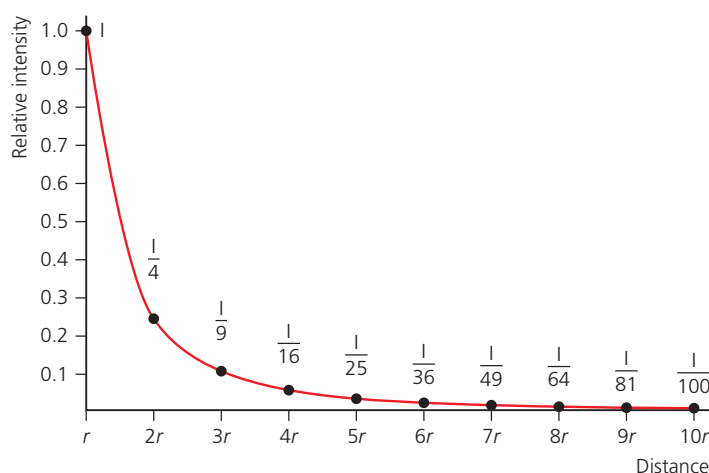
$$\text{intensity, } I \propto \frac{1}{\text{distance}^2}$$

Figure C3.8 represents this kind of relationship in graphical form.

But note that, as before, if the waves are a great distance from their point source, the wavefronts will become almost parallel / plane waves, so that no loss of energy/power due to spreading may be noticeable.



■ **Figure C3.7** Amplitude of circular wavefronts decreases with the distance from source



■ **Figure C3.8** Inverse square relationship

Tool 3: Mathematics

Plot linear and non-linear graphs showing the relationship between two variables

Check the data shown in Table C3.1 to determine if there is an inverse square relationship between x and y :

- numerically
- graphically.

■ Table C3.1

x	y
1.34	9
0.96	17
0.81	24
0.70	32
0.64	38
0.59	45
0.55	52
0.52	58
0.49	66

- 4 Light and infrared radiation arriving perpendicularly on a solar panel have a total intensity of 780 W m^{-2} .
 - a If the panel has dimensions of $50 \times 80 \text{ cm}$, calculate the incident power.
 - b Explain why this power will change during the course of the day.
- 5 A girl is reading a book at night using the light from a single lamp, which may be assumed to be a point source. If the lamp was originally 1.80 m away from the book, show that the intensity of the light on the book doubles if it is moved 0.53 m closer to the lamp.
- 6 Figure C3.9 shows some typical aerials used for transmitting (and receiving) signals to mobile phones.



■ Figure C3.9 Typical aerials for transmitting mobile phone signals

- a Explain why it is desirable that the waves from these aerials are *not* emitted equally in all directions.
 - b Suggest which feature of the aerials limits the vertical spreading of the waves.
- 7 The radiation from the Sun which reaches the top of the Earth's atmosphere has an intensity of 1360 W m^{-2} (see Topic B.1). It is approximately 40% visible light, 50% infrared and 10% ultraviolet. The intensity reaching the Earth's surface is approximately 1000 W m^{-2} . The approximate proportions reaching the Earth's surface are 44% visible light, 53% infrared and 3% ultraviolet. Use this data to estimate the percentages of these three radiations which are:
 - a transmitted by the Earth's atmosphere
 - b absorbed / scattered by the Earth's atmosphere.
- 8 Why is the sky blue? (Research on the internet if necessary.)
- 9 It is considered to be a health risk to expose our ears to sounds of intensity greater than 10 mW m^{-2} for more than a few minutes.
 - a Calculate the total power received on an eardrum of area 0.48 cm^2 from this intensity.
 - b If the sound intensity 2.10 m from a loudspeaker at a rock concert was 0.44 W m^{-2} , estimate how far away you would need to be in order to reduce the intensity to 100 mW m^{-2} .
 - c State two assumptions that you made in answering b. Discuss whether these assumptions are reasonable.